

9. (a) Food poisoning is a notifiable disease in Wales. Doctors are required to report every case of food poisoning.

(i) Explain how some bacteria cause food poisoning. [2]

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(ii) State **two** symptoms of food poisoning. [1]

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(iii) State **two** different types of precaution that can be taken when **preparing** food to reduce the risk of food poisoning. [2]

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- (b) **Figure 1** shows the number and rate of reported cases of food poisoning in Wales between 1992 and 2016.

**Figure 1**

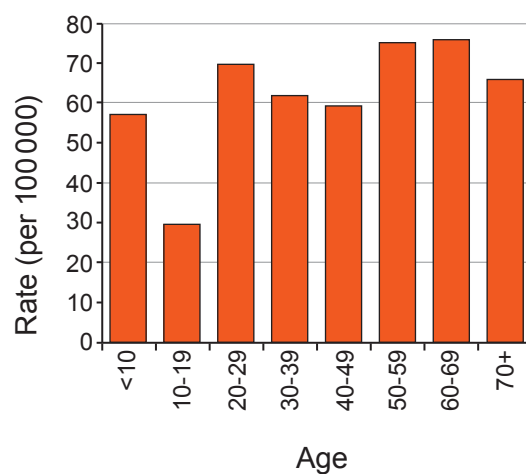
| Year | Total number of reported cases | Rate (number per 100 000 population) |
|------|--------------------------------|--------------------------------------|
| 1992 | 3 590                          | 124.5                                |
| 1998 | 5 946                          | 205.0                                |
| 2000 | 4 716                          | 162.2                                |
| 2006 | 4 301                          | 145.4                                |
| 2010 | 4 980                          | 165.6                                |
| 2014 | 4 516                          | 146.1                                |
| 2016 | 4 504                          | 145.7                                |

The percentage of reported cases of food poisoning during each quarter of a year is shown in **Figure 2**.

**Figure 2**

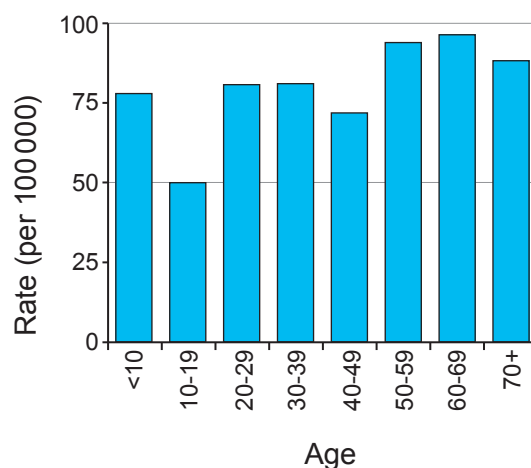
| Quarter | % of total for year |      |      |
|---------|---------------------|------|------|
|         | 2012                | 2014 | 2016 |
| Jan-Mar | 18                  | 17   | 19   |
| Apr-Jun | 26                  | 26   | 25   |
| Jul-Sep | 32                  | 32   | 33   |
| Oct-Dec | 24                  | 25   | 23   |

**Figure 3** shows the rate of food poisoning amongst females by age group in 2016.

**Figure 3**

**Figure 4** shows the rate of food poisoning amongst males by age group in 2016.

**Figure 4**



Use the data in **Figures 1 to 4** to answer the following questions.

- (i) Since 1998, egg-laying hens have been vaccinated against a species of *Salmonella*. Explain how this has affected the number of reported cases of food poisoning. [2]

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- (ii) Explain the variation in the percentage of reported cases of food poisoning in each of the quarters in a year. [3]

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- (iii) It is suggested that the rate of food poisoning does not depend on gender or age. Explain whether this suggestion is confirmed by the data. [2]

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**END OF PAPER**